Weakly-Supervised Domain Adaptation for Cyber Threat Intelligence Named Entity Recognition: A Comparison of Span-Matching and Token-Classification Architectures

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Abstract

Cyber threat intelligence (CTI) reports contain entity types with no general-domain analogue — ATT&CK technique identifiers, file hashes, TLP markings, threat actor aliases — and zero-shot named entity recognition (NER) models perform poorly on them in systematic, structural ways. We fine-tune GLiNER, a zero-shot span-matching NER model, on a silver corpus derived from ensemble agreement between a high-precision dictionary annotator (STIX indicators, MITRE ATT&CK, IOC regexes) and zero-shot GLiNER itself, requiring no human-labeled training data. On a held-out, document-level test split of CISA advisories, fine-tuning raises micro-F1 from 0.20 to 0.67, with the largest gains on structured types (technique IDs 0.24→0.99, file hashes 0.04→0.66). We further compare this span-matching architecture against a directly comparable token-classification baseline (RoBERTa) fine-tuned on the identical weak labels and scored with the identical evaluation harness, and against a frontier LLM (Gemini) run zero-shot on full documents. **[PENDING: finalize abstract numbers once RoBERTa-large/SecureBERT and the gold audit complete]**.

1 Introduction

Cyber threat intelligence reports — government advisories, malware analyses, incident write-ups — are the primary vehicle by which defenders share knowledge, yet extracting structured information from them remains largely manual. Named entity recognition over CTI text must handle entity types with no general-domain analogue: ATT&CK technique identifiers, file hashes, TLP markings, YARA rule names, threat actor aliases. Modern zero-shot span-matching models such as GLiNER (Zaratiana et al., 2024) promise NER without task-specific training data, but their performance on security text is poor in systematic, structural ways.

Human annotation at scale is expensive in this domain because it requires security expertise. This motivates our central question: can weak supervision derived from *ensemble agreement* between independent, imperfect annotators close the domain gap without any human labeling? We build a silver-labeled corpus from agreement between a dictionary annotator (structured, high-precision on IOCs) and zero-shot GLiNER (semantic, higher-recall on contextual entities), fine-tune GLiNER on the result, and evaluate rigorously against a held-out, document-level test split.

We ask four research questions:

RQ1 Does ensemble-agreement weak supervision close the domain gap between zero-shot and fine-tuned GLiNER?

RQ2 Which entity types benefit most from such fine-tuning, and does the answer depend on model architecture (span-matching vs. token-classification) under the identical weak labels?

RQ3 Do the base model's documented failure modes — confusing technique IDs with file hashes, tagging defender organizations as threat actors — survive fine-tuning?

RQ4 How does fine-tuned GLiNER's quality/latency/cost trade-off compare to a frontier LLM run zero-shot on the same task?

Our contribution extends the original weak-supervision result with a controlled architecture comparison: we fine-tune RoBERTa (Liu et al., 2019), a token-classification model with no zero-shot span-matching prior, on the *identical* silver labels, and score it with the identical document-level evaluation harness used for GLiNER. This isolates architecture from data — both models see the same weak supervision, so any systematic difference in what they learn (e.g. which label types they refuse to predict) is attributable to the modeling approach, not the training signal.

2 Related Work

GLiNER (Zaratiana et al., 2024) treats NER as matching natural-language entity-type descriptions against learned span embeddings, enabling zero-shot extraction of arbitrary types; its schema-driven successor GLiNER2 (Zaratiana et al., 2025) unifies NER with classification and structured extraction in one model. Distant supervision for NER (Devlin et al., 2019) was formalized for open-domain text by BOND (Liang et al., 2020), which fine-tunes a token classifier on knowledge-base-derived labels then self-trains through a teacher-student loop to denoise them — the same failure modes we target with ensemble agreement (incomplete and noisy weak labels) motivate BOND's two-stage design.

Cybersecurity-specific NER corpora suffer acutely from schema fragmentation: Schmidt et al. (2025) show that naively unifying four CTI datasets under a coarse shared label set *hurts* performance relative to single-dataset training, while Ech-Chammakhy et al. (2025) demonstrate that principled, STIX-anchored harmonization recovers a 30% relative F1 gain over the same naive union — evidence that *how* labels are unified matters as much as how much data is unified. Deka et al. (2024) extend the entity-type inventory to attribution-specific concepts (impact, course-of-action) with deliberately long, sentence-level spans, and find instruction-fine-tuned LLMs outperform encoder taggers on their benchmark — the inverse of the pattern this paper reports on our IOC-heavy corpus, suggesting the LLM-vs-encoder trade-off is span-length- and entity-type-dependent rather than universal. Alam et al. (2024) benchmark LLMs directly on CTI tasks including entity extraction, providing the frontier-model baseline methodology RQ4 adapts. A fuller treatment of the method landscape appears in our companion survey (`paper/survey.md`, this repository).

3 Data and Method

3.1 Pipeline and Silver Corpus

We scrape approximately 5,000 CISA cybersecurity advisories and apply a three-layer weak-supervision ensemble to 313 advisory PDFs (converted to markdown via docling): (1) a dictionary/regex layer over STIX indicator bundles and IOC patterns, (2) a MITRE ATT&CK lookup (~2,400 entities: groups, tools, malware, technique IDs), and (3) zero-shot GLiNER itself, run over the same text. Training supervision is the *agreement* between the dictionary layer and GLiNER's zero-shot predictions, plus dictionary-only spans forced positive as hard negatives so the model learns structured patterns the base model's zero-shot pass already missed. This yields 16 CTI-specific labels across **[PENDING: confirm final chunk/entity counts from create_training_data.py –stats]** chunks from 82 densely-annotated advisories.

3.2 Split

We split by advisory *document*, not by chunk, before chunking for training — 66 advisories train / 8 validation / 8 test — so that overlapping text windows from the same document cannot appear on both sides of the split. This was a deliberate fix: chunk-level splitting leaks document context (shared boilerplate, repeated entity mentions) across the train/test boundary and inflates apparent performance.

3.3 Models

Zero-shot GLiNER. `urchade/gliner_medium-v2.1`, run with no fine-tuning, as the baseline RQ1 measures against.

Fine-tuned GLiNER. The same checkpoint, fine-tuned on the silver corpus (5 epochs, LR 5e-5, effective batch 32).

RoBERTa (this paper's architecture-comparison contribution). `roberta-base` (Liu et al., 2019), fine-tuned as a standard BIO token classifier on the identical silver labels. Because gold spans are word-level and RoBERTa's tokenizer is subword-based, we tag the first wordpiece of each word and mask continuation wordpieces from the loss. Corpus chunks range from 22 to 292 words, and long or hash-heavy chunks fragment into more wordpieces than a single 512-token window admits under naive truncation; we instead tokenize with overlapping sliding windows (~510-wordpiece budget, adaptive overlap) so every word is covered by at least one window, and take each word's predicted tag from the window in which it sits most centrally at inference time. We further train `roberta-large` and SecureBERT (Aghaei et al., 2022) — a RoBERTa-architecture model domain-pretrained on security text — under the identical recipe, to separate the effect of model scale from domain-pretraining. **[PENDING:**

roberta-large / SecureBERT results pending training completion]

Frontier LLM (RQ4). Gemini 3.5 Flash, run zero-shot with no fine-tuning, on full documents via chunked inference (15,000-character windows, 500-character overlap, entities merged across chunk boundaries).

3.4 Evaluation

All models are scored with the same document-level harness: predictions from overlapping chunks are merged back to document-global token indices before matching, so entities in chunk-overlap regions are not double-counted. We report strict (exact span-and-label match) and relaxed (label match plus span overlap) micro/macro precision, recall, and F1. Crucially, RoBERTa and GLiNER are scored with the *same* scoring implementation (`scraper/evaluate.py`) so the comparison in Section 4 is not an artifact of different matching rules.

4 Results

4.1 RQ1/RQ2: Weak Supervision Closes the Gap, Unevenly

Table 1 shows zero-shot GLiNER improving from 0.20 to 0.67 micro-F1 after fine-tuning on the silver corpus, with the largest absolute gains concentrated in structured, dictionary-friendly types — confirming RQ2's hypothesis. Technique IDs and file hashes, where zero-shot GLiNER is near-unusable (F1 0.24 and 0.04, precision as low as 0.02 for hashes), reach 0.99 and 0.66 after fine-tuning. Semantic types with sparser or noisier weak supervision (tactic, threat actor group, attack technique name) improve but remain the weakest categories (F1 0.13–0.36).

4.2 RQ2 (continued): Architecture Comparison Under Identical Weak Labels

Table 2 compares fine-tuned GLiNER against `roberta-base`, trained on the *identical* silver corpus and scored with the identical harness. RoBERTa achieves a higher overall micro-F1 (0.74 vs. 0.67) but the aggregate number conceals an important qualitative difference: RoBERTa is precision-heavy (0.91 vs. 0.62) and recall-light (0.62 vs. 0.71), and — most notably — predicts *zero* entities of four types GLiNER at least attempts: *tactic*, *threat actor group*, *file name*, and *attack technique name*, all types with fewer than 50 training examples. We attribute this to architectural differences in how each model handles class imbalance: a softmax classification head must learn a discriminative weight vector per label and, under severe imbalance, can learn to never fire on rare classes; GLiNER's span-matching mechanism instead embeds label *names* into the same representation space as spans, which generalizes to few-shot labels rather than requiring per-label discriminative capacity. This is consistent with GLiNER's original zero-shot motivation (Zaratiana et al., 2024) and suggests the architectural choice matters independently of the fine-tuning data — answering RQ2's architecture-dependence question in the affirmative. **[PENDING: roberta-large and SecureBERT rows once training completes — tests whether scale or domain pretraining recovers the zero-prediction labels]**

4.3 RQ4: Comparison to a Frontier LLM

Table 3 compares fine-tuned GLiNER against Gemini 3.5 Flash run zero-shot on full documents. Fine-tuned GLiNER outperforms Gemini by 11 F1 points (0.67 vs. 0.56) while being approximately 600× faster at inference (<0.1s vs. 66s per document) and free to run locally. Gemini's per-label profile shows it is competitive on structured patterns (TLP 0.97, technique IDs 0.93, CVE IDs 0.92) — these are not a frontier-model weakness — but weak on file hashes (0.48) and tactics (0.00), and its precision on threat actor group (0.18 at 0.92 recall) suggests many of its "false positives" are entities our dictionary gold labels miss rather than genuine model errors (see hand-audit notes, Section 6).

5 Analysis: Do the Failure Modes Survive Fine-Tuning? (RQ3)

[PENDING: Targeted failure-mode analysis pending — probes are being run against raw per-span predictions (not just aggregate metrics) for both zero-shot and fine-tuned GLiNER on the test set, searching specifically for (a) technique-ID/file-hash label confusion and (b) defender organizations tagged as threat actors. Results and concrete examples will be inserted here.]

6 Limitations

Circular silver evaluation. Our test-set gold labels are themselves silver (dictionary/MITRE-

Label	Zero-shot P	R	F1	Fine-tuned P	R	F1	n
malware family	0.55	0.29	0.38	0.94	0.56	0.70	575
tlp classification	0.70	0.19	0.30	0.93	0.65	0.76	335
attack technique id	0.42	0.17	0.24	0.99	1.00	0.99	295
file hash	0.02	0.08	0.04	0.53	0.90	0.66	79
cve identifier	0.56	0.94	0.70	0.85	1.00	0.92	50
tactic	0.00	0.00	0.00	0.16	0.85	0.27	47
threat actor group	0.03	0.48	0.06	0.25	0.60	0.36	25
file name	0.10	0.47	0.16	0.12	0.60	0.21	15
attack technique name	0.05	0.08	0.06	0.08	0.42	0.13	12
hacking tool	0.01	0.25	0.01	0.06	0.62	0.10	8
cwe identifier	0.00	0.00	0.00	0.25	1.00	0.40	1
Micro	0.17	0.24	0.20	0.62	0.71	**0.67**	1442

Table 1: Zero-shot vs. fine-tuned GLiNER, strict span+label match, full-document test set (8 held-out advisories, 1,442 gold spans). Labels with zero support in the test set omitted.

Model	P	R	F1	Macro F1
GLiNER (fine-tuned)	0.62	**0.71**	0.67	0.34
RoBERTa-base (fine-tuned)	**0.91**	0.62	**0.74**	**0.46**
[PENDING: RoBERTa-large]				
[PENDING: SecureBERT]				

Table 2: Architecture comparison, identical silver labels, identical evaluation harness. Strict span+label, full-document test set.

Model	F1	Latency	Cost
GLiNER (fine-tuned)	**0.67**	<0.1s	free
Gemini 3.5 Flash	0.56	66s	~$0.0001/doc
GLiNER (zero-shot)	0.20	<0.1s	free
GLiNER2 (zero-shot)	0.07	6.3s	free

Table 3: Full-document, apples-to-apples comparison. Gemini via Vertex AI, thinking disabled, chunked inference merged across chunk boundaries.

derived), and fine-tuned GLiNER is trained toward agreement with a closely related signal, so evaluating against silver labels risks overstating quality where the model has learned to reproduce silver-label *noise* rather than genuine entities. We identify a concrete instance: the PDF footer artifact "Page" is silver-annotated as a malware family 160 times, and the fine-tuned model reproduces this error. **[PENDING: Gold audit — a blind, hand-judged sample of ~300 predicted spans across models — is in progress and will report honest precision independent of silver label noise; results pending.]**

Small test set. Eight held-out advisories (1,442 gold spans) is small enough that five labels have zero test-set support and several more have fewer than 15 examples; per-label numbers for rare types should be read as indicative, not precise.

Windowing artifacts in the RoBERTa comparison. RoBERTa's sliding-window inference resolves each word's tag from whichever window contains it most centrally, which is a heuristic, not a principled ensembling method; a small fraction of predictions near window boundaries may be affected.

7 Conclusion

[PENDING: Final conclusion pending completion of RQ2's architecture extension (RoBERTa-large, SecureBERT), RQ3's failure-mode probes, and the gold audit.]

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